Faculty Profile

Dr. Piyush Bagla is a dedicated academician and researcher with a Ph.D. degree in Computer Science and Engineering, specializing in Machine Learning and Natural Language Processing from Dr. B.R. Ambedkar National Institute of Technology (NIT), Jalandhar, India. Dr. Bagla believes that effective teaching goes beyond theoretical instruction involves connecting concepts to real-world applications that resonate with students. For more detail, kindly visit the link below.

Profile Link - https://www.upes.ac.in/faculty/school-of-computer-science/dr-piyush-bagla

Java Programming

History of Java : Part 1

James Gosling, Mike Sheridan, and Patrick Naughton initiated the Java language project in June 1991.
The small team of sun microsystems engineers called **Green Team**.

Originally designed for small, embedded systems in electronic appliances like set-top boxes.

Firstly, it was called "**Greentalk**" by James Gosling and file extension was .gt.

After that, it was called **Oak**, and then finally, **JAVA**.

History of Java : Part 2

Java is an island of Indonesia where first coffee was produced (called java coffee).

Originally developed by James Gosling at Sun Microsystems (which is now a subsidiary of Oracle Corporation) and released in 1995

In 2010 the company was purchased by Oracle Corporation

Notice that Java is just a name, not an acronym

JDK Evolutions

23 Jan. 1996
JDK 1.0

JDK 1.1
19 Feb. 1997

8 Dec. 1998
J2SE 1.2

J2SE 1.3
8 May 2000

6 Feb. 2002
J2SE 1.4

J2SE 5.0
30 Sep. 2004

11 Dec. 2006
Java SE 6

Java JDK 24.0.2
19 Sep. 2025

The Story of Java: Problems, Solutions, and the Internet

1. What was the problem with other languages?

A. Programs didn't run everywhere

- Imagine writing a letter in Microsoft Word, but you can only open it on *your* laptop, not on your friend's.
- That was how C and C++ worked. If you wrote a program on Windows, you had to re-write or adjust it for Linux or Mac.

How Java fixed it:

Java said: *"Write once, run anywhere."*

It introduced the **Java Virtual Machine (JVM)**—like a universal translator.

- You write code in Java.
- It gets translated into a common language (called bytecode).
- Any computer with JVM can run it, no matter the operating system.

👉 Real-life example: A bank could write one Java program and run it on all their ATMs, no matter the machine brand.

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B. Memory problems and crashes

- In C/C++, programmers had to manually manage memory. Imagine packing a suitcase but also having to remember to remove each unused item at the right time—if you forget, your bag overflows.
- This caused crashes, leaks, and even hackers breaking into systems.

How Java fixed it:

Java has a **garbage collector**. It's like a cleaning robot that automatically removes unused items from your suitcase.

- Safer
- Fewer crashes
- Less work for the programmer

👉 Real-life example: Telecom companies used Java for phone systems, which needed to run for months without crashing. The garbage collector helped.

The Story of Java: Problems, Solutions, and the Internet

2. What was wrong with the Internet—and how Java helped

Problem 1: The Web was static

- In the 90s, websites were like posters—just text and images. No videos, no interactive features.

Java's role:

Java Applets (small programs inside websites) made pages interactive: animations, calculators, stock tickers.

👉 Example: Early online classrooms used Java applets to run whiteboards and quizzes.

The Story of Java: Problems, Solutions, and the Internet

Problem 2: Security fears

- People were afraid of downloading programs because they could be viruses.
- Windows programs had too much power—they could read your files or install malware.

Java's role:

Java introduced the **sandbox**: downloaded code could only run in a restricted area.

- Couldn't delete your files.
- Couldn't access your data without permission.

👉 Example: Online banking used Java for secure login tools.

The Internet and Cross-Platform Problem

In the early days of the Internet (mid-1990s):

- A company that wanted to put its **software online** had users on **different types of computers**.
  - Some on **Windows** (most offices).
  - Some on **Mac** (designers, universities).
  - Some on **Linux/UNIX/Solaris** (servers, researchers).
- If you wrote your program in **C or C++**, you had to create **separate versions** for each operating system.
  - Expensive 💰
  - Time-consuming ⏳
  - Error-prone 🐞

The Internet and Cross-Platform Problem

☕ How Java Fixed This for the Internet

Java introduced the **JVM (Java Virtual Machine)**.

- You wrote your code once in Java.
- It was converted to **bytecode**.
- That bytecode could run on **any machine with a JVM installed** (Windows, Mac, Linux, Solaris, etc.).

This was huge for the Internet because:

1. A company could put one Java program (like an applet or a server app) online.
2. Any user, regardless of their computer type, could access it.

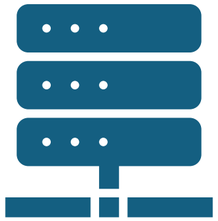
What is java?

- A general-purpose object-oriented language.
- Write Once Run Anywhere (WORA).
- Designed for easy Web/Internet applications.
- Widespread acceptance.

Java Editions

J2SE(Java 2 Standard Edition) - to develop client-side standalone applications or applets.

J2ME(Java 2 Micro Edition) - to develop applications for mobile devices such as cell phones.



J2EE(Java 2 Enterprise Edition) - to develop applications server-side such as Java servlets and Java ServerPages